



SYMBIOSIS INSTITUTE OF TECHNOLOGY (SIT)

Constituent of Symbiosis International (Deemed University), Pune

(Established under Section 3 of the UGC Act of 1956 vide notification number F-9-12/2001-U-3 of the Government of India)
Re-Accredited by NAAC with 'A++' Grade

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Class : CSE
Batch : 2025-29

Assignment : 16

Subject: Programming with Java

ASSIGNMENT No : 16

TITLE : Write a Java program to create and handle a user-defined exception.

Problem Statement

Write a Java program to create and handle a user-defined exception.

Learning Objectives

To create and implement custom exceptions for application-specific error handling.

Theory

User-defined exceptions are created by extending the Exception class. They allow developers to define meaningful exceptions for specific business logic. Custom exceptions improve program readability and simplify debugging by providing descriptive error messages. They are widely used in enterprise software and validation systems.



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Code and Output Screenshots

```
Exp16.java > Java > Exp16
1  class MyException extends Exception {
2      MyException(String message) {
3          super(message);
4      }
5  }
6  public class Exp16 {
7      Run main | Debug main | Run | Debug
8      public static void main(String[] args) {
9          try {
10             throw new MyException("Something went wrong");
11         }
12         catch (MyException e) {
13             System.out.println(e.getMessage());
14         }
15     }
16 }
```

Problems 39 Output Debug Console Terminal Ports

- PS C:\Users\LENOVO\Documents\JAVA Assignments> javac Exp16.java
- PS C:\Users\LENOVO\Documents\JAVA Assignments> java Exp16
- Something went wrong
- PS C:\Users\LENOVO\Documents\JAVA Assignments>



Conclusion

Thus, the program successfully demonstrates the creation and handling of a custom exception in Java using try, throw, and catch. The custom exception is thrown with a message and handled using catch, displaying the error message without terminating the program abruptly.

Exercise

Q1) Create a Voting System that throws a custom exception if age is below 18

Sol) Code :

```
J Excp16.java > Java > Excp16
1  import java.util.Scanner;
2  class InvalidAgeException extends Exception {
3      InvalidAgeException(String message) {
4          super(message);
5      }
6  }
7  public class Excp16 {
8      Run main | Debug main | Run | Debug
9      public static void main(String[] args) {
10         Scanner sc = new Scanner(System.in);
11         System.out.print("Enter age: ");
12         int age = sc.nextInt();
13         try {
14             if (age < 18) {
15                 throw new InvalidAgeException(
16                     "Age below 18. Applicant not eligible to vote."
17                 );
18             }
19             System.out.println("Applicant Eligible to vote.");
20         }
21         catch (InvalidAgeException e) {
22             System.out.println(e.getMessage());
23         }
24     }
25 }
```



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Output :

Case 1 : age < 18

```
Problems 40 Output Debug Console Terminal Ports
PS C:\Users\LENOVO\Documents\JAVA Assignments> cd "c:\Users\LENOVO\Documents\JAVA Assignments\" ; if ($?) { javac Excp16.java } ; if ($?) { java Excp16 }
Enter age: 15
Age below 18. Applicant not eligible to vote.
PS C:\Users\LENOVO\Documents\JAVA Assignments> 
```

Case 2 : age > 18

```
Problems 40 Output Debug Console Terminal Ports
PS C:\Users\LENOVO\Documents\JAVA Assignments> cd "c:\Users\LENOVO\Documents\JAVA Assignments\" ; if ($?) { javac Excp16.java } ; if ($?) { java Excp16 }
Enter age: 21
Applicant Eligible to vote.
PS C:\Users\LENOVO\Documents\JAVA Assignments> 
```



Q2) Create a Driving License System that throws a custom exception if the user's age is below 18. If the age is valid, display that the user is eligible for a driving license.

Sol) Code :

```
1  import java.util.Scanner;
2
3  class InvalidAgeException extends Exception {
4      InvalidAgeException(String message) {
5          super(message);
6      }
7  }
8  public class Excp162 {
9      Run main | Debug main | Run | Debug
10     public static void main(String[] args) {
11         Scanner sc = new Scanner(System.in);
12         System.out.print("Enter age: ");
13         int age = sc.nextInt();
14
15         try {
16             if (age < 18) {
17                 throw new InvalidAgeException("Age is below 18. Applicant not eligible for driving license.");
18             }
19
20             System.out.println("Applicant eligible for driving license.");
21         }
22         catch (InvalidAgeException e) {
23             System.out.println(e.getMessage());
24         }
25     }
26 }
```

Output :

Case 1 :

```
Problems 41 Output Debug Console Terminal Ports
PS C:\Users\LENOVO\Documents\JAVA Assignments> cd "c:\Users\LENOVO\Documents\JAVA Assignments\" ; if ($?) { javac Excp162.java }
; if ($?) { java Excp162 }
Enter age: 13
Age is below 18. Applicant not eligible for driving license.
PS C:\Users\LENOVO\Documents\JAVA Assignments>
```

Case 2 :

```
Problems 41 Output Debug Console Terminal Ports
PS C:\Users\LENOVO\Documents\JAVA Assignments> cd "c:\Users\LENOVO\Documents\JAVA Assignments\" ; if ($?) { javac Excp162.java }
; if ($?) { java Excp162 }
Enter age: 24
Applicant eligible for driving license.
PS C:\Users\LENOVO\Documents\JAVA Assignments>
```